

Advancing Managed Retreat in California

Planning Considerations & Buyout-Leaseback Financing Evaluation

William Dean, Lilia Mourier, Wesley Noble, Ada Olumba, & Daniel O'Shea
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UC SANTA BARBARA

Bren School of Environmental
Science & Management

UCSB Bren School, Managed Shores GP | Mourier, L., O'Shea, D., Noble, W., Dean, W., & Olumba, A.

Our Team



**Ada
Olumba**



**Daniel
O'Shea**



**Lilia
Mourier**



**William
Dean**



**Wesley
Noble**

Acknowledgements

Project Advisors

Dr. Andrew Plantinga, Faculty Advisor, UCSB Bren School

Jonah Danzinger, PhD Advisor, UCSB Economics

Summer Gray, External Advisor, UCSB Environmental Studies

Kim Kimbell, External Advisor, UCSB Bren School Dean's Council

Client Partners

Dr. Charles Lester, Director, UCSB Ocean and Coastal Policy Center

Carey Batha, Environmental Scientist, California Coastal Commission

Kelsey Ducklow, Coastal Resilience Coordinator, California Coastal Commission

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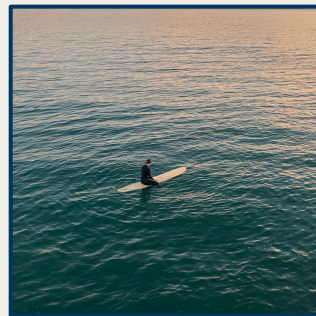
JAMES S. BOWER
FOUNDATION

Re-envisioning.

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More Than a Shoreline

- Globally-significant **coastal habitats** & **infrastructure**
- California **coastal property is among the most valuable in the world**
- Supports **\$44 billion-dollar coastal economy**
- Used for **recreation, cultural** and **spiritual well-being**, a connection to nature, and to support **livelihoods**.



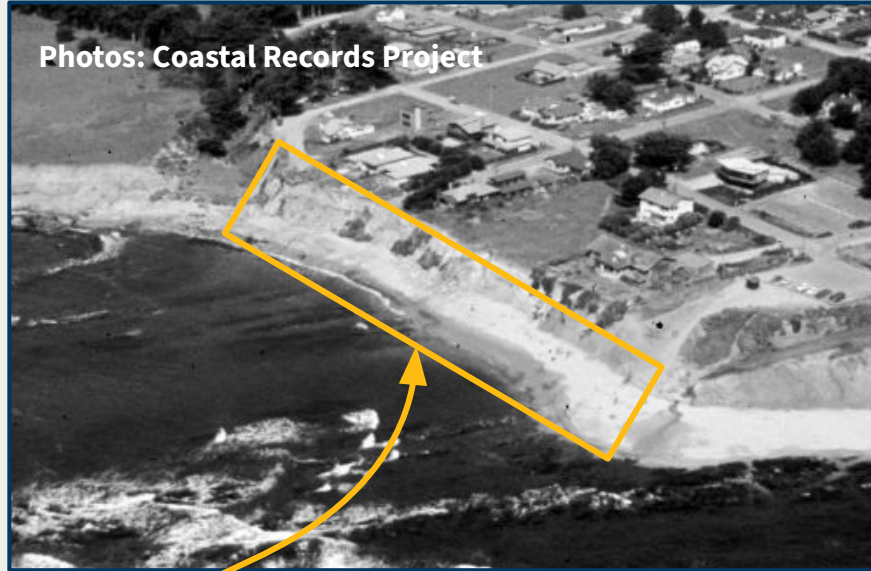
Threats From Rising Seas

Climate change is accelerating sea level rise and intensifying storms, putting California's coast at greater risk of flooding and erosion.



Photos: California King Tides Project, California Coastal Commission

Hard Protection and The Coastal Squeeze



1972

Beach is present, development is minimal, **no hard protection**

Hard protection made of rock revetment



2024

Beach only remains **where there is space for inland transgression**

From Protection to Relocation

Managed Retreat

The planned relocation or removal of people and infrastructure away from areas increasingly vulnerable to coastal hazards.

Buyout-Leaseback

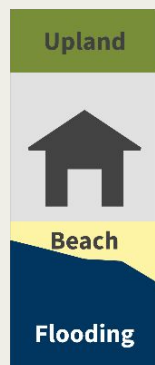
One managed retreat strategy in which a public agency purchases a vulnerable property and leases it back to the previous owner or other occupant(s) for a limited time before retreat occurs.

Buyout-Leaseback Program Phases



1.

Local government purchases property at a **buyout price** well before retreat is necessary.



2.

Local government leases property back to owner or new occupant. **Rental revenue** repays buyout loan over **time**.



3.

Local government ends lease based on predefined **safety, financial, or environmental** triggers.



4.

Local government **demolishes** structure and **restores** land.



5.

Coastal habitats **recover and migrate** inland as sea level rises.

Recent Legislative Momentum



OFFICE OF THE GOVERNOR

SEP 29 2022

To the Members of the California State Senate:

I am returning Senate Bill 1078 without my signature.

This bill requires the Ocean Protection Council to develop the Sea Level Rise Revolving Loan Pilot Program to provide low-interest loans to local jurisdictions to purchase identified vulnerable coastal properties located in communities and populations disproportionately affected by climate change.

Unfortunately, this bill does not comprehensively address the cost, likely to be carried out over decades.

Climate-driven sea level rise presents major land-use planning challenges. The scope and scale of the problem, and the work necessary to make sure that California can adapt to rising seas, requires innovative planning approaches and implementation action at the state and local level. Financial tools, such as the one proposed in SB 1078, have the potential to play an important role in a portfolio of strategies that will help build coastal resilience in California.

Such an effort should be considered within a comprehensive lens that:

1) encourages cooperation and feasibility among all necessary stakeholders; 2) considers the impacts of inserting local jurisdictions in the property rental market; and 3) allows for unified agreement by local, state, and federal partners that the type of policy tool in SB 1078 is the best path forward for coastal residential properties.

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“Allows unified agreement by local, state, and federal partners that [buyout-leaseback] is the best path forward for coastal residential properties

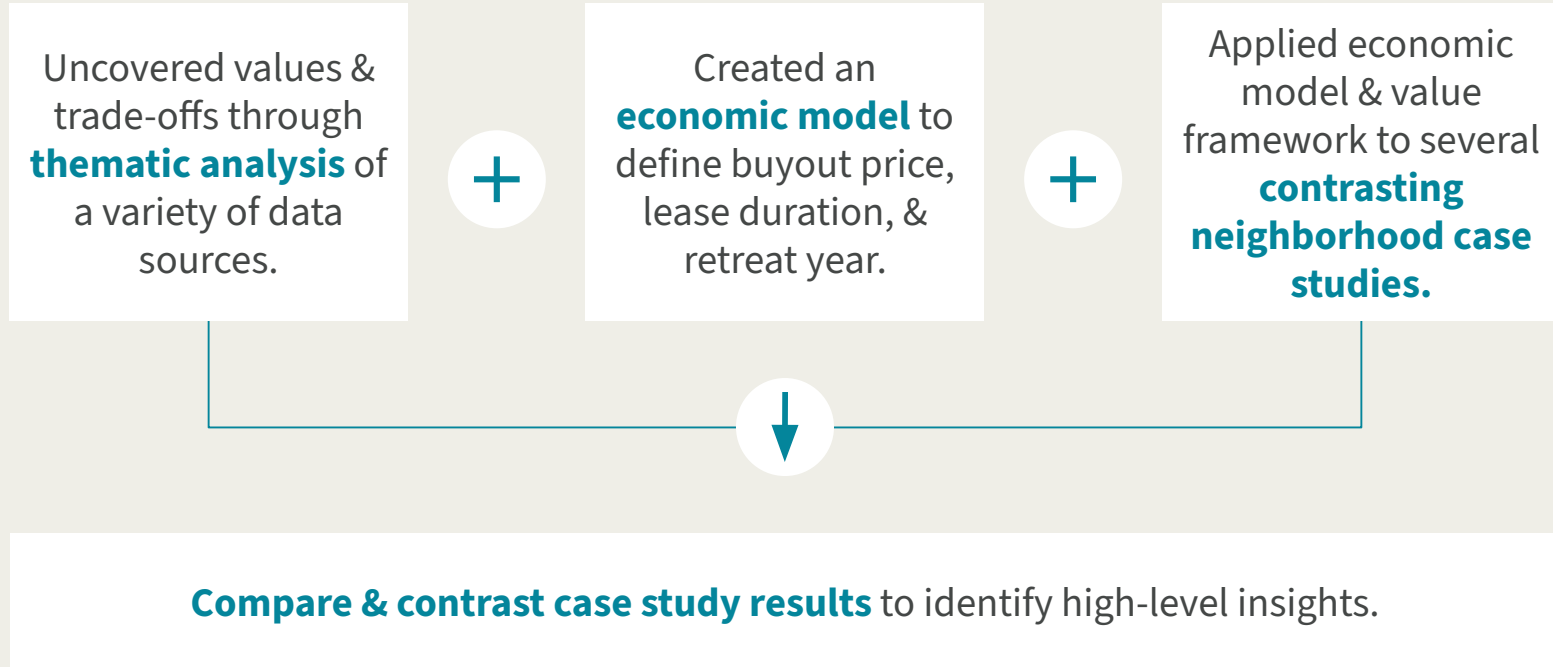
Key Project Questions

1 What values, trade-offs, and challenges shape managed retreat planning decisions across California?

2 When does occupation of a coastal residential property become economically unsustainable?

3 How can this timing inform proactive acquisition strategies that balance public cost with fair compensation to property owners?

Our Approach



From Sources to Themes

150+ Sources

700+ Codes

80+ Themes

Journal
Articles

"When residents learn about plans from a **trusted local source** who supports the changes, they are **more likely to be accepting**. As more people support the plan, the effect snowballs..." - Bragg (2021)

Legal
Documents

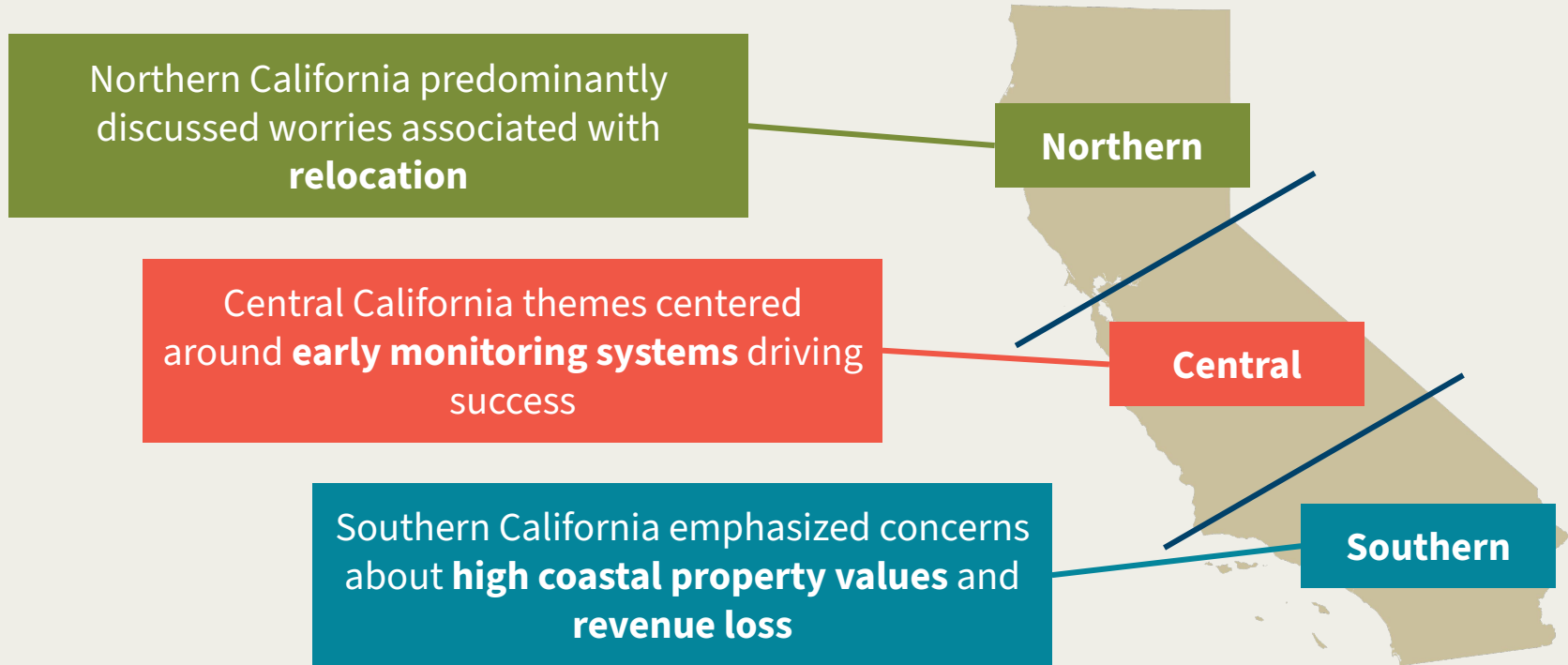
Government
Reports

"Communities must decide whether to take a **proactive approach** that is controlled and optimized to maximize public and private benefits, or a **reactive** one characterized by **sudden, emergency responses** to extreme events like landslides and cliff collapse." — San Mateo County Sea Level Rise Vulnerability Assessment, 2022

News & Public
Comments

"Del Martians will deal with it [Sea Level Rise] **when they see the actual impacts of global warming** on a more weekly, yearly basis," he said. "And **until that happens, it's still theoretical**." - John, 2019

Regional Emphases



Top Cross-Cutting Themes



Inclusive engagement, equity, and public **stewardship** are foundational to retreat success.



Funding gaps, property value disparities, and **local revenue concerns** complicate implementation.



Coordination across jurisdictions, agencies, and planning scales is essential.



Communication, distrust, and **fears of displacement** strongly shape public acceptance.



Managed retreat can deliver **major environmental** and **long-term economic benefits** when planned proactively.

Buyout-Leaseback Economic Model

1. Estimate a property's remaining economic lifespan

Flood Hazard Properties

Retreat is triggered when future flood damages outweigh future rental value.

Cliff Erosion Hazard Properties

Retreat is triggered when the projected cliff edge comes within 10 meters of the structure.

Buyout-Leaseback Economic Model

1. Estimate a property's remaining economic lifespan

2. Set the property buyout price based on remaining use value

The buyout reflects the value the property can still provide before retreat is needed, not simply today's full market price.

Buyout-Leaseback Economic Model

1. Estimate a property's remaining economic lifespan

2. Set the property buyout price based on remaining use value.

3. Calculate leaseback revenue and flood damages

Estimate rental income after acquisition and expected damage costs through the retreat year.

Buyout-Leaseback Economic Model

1. Estimate a property's remaining economic lifespan

2. Set the property buyout price based on remaining use value.

3. Calculate leaseback revenue and flood damages

4. Calculate net public cost

Compare buyout cost, repair costs, and leaseback revenue at the property and neighborhood scale.

Five Neighborhood Case Studies



Comparing Retreat Timing and Public Cost

Case Study	Hazard	Median Retreat Year	Median Property Cost	Total Neighborhood Cost
King Salmon	Flooding	22	\$110 K	\$6.5 M
Stinson	Flooding	32	\$660 K	\$132 M
Pacifica	Cliff Erosion	38	-\$550 K	-\$36 M
Isla Vista	Cliff Erosion	34	-\$664	-\$24 M
Carpinteria	Flooding	22	\$530 K	\$75 M

Comparing Retreat Timing and Public Cost

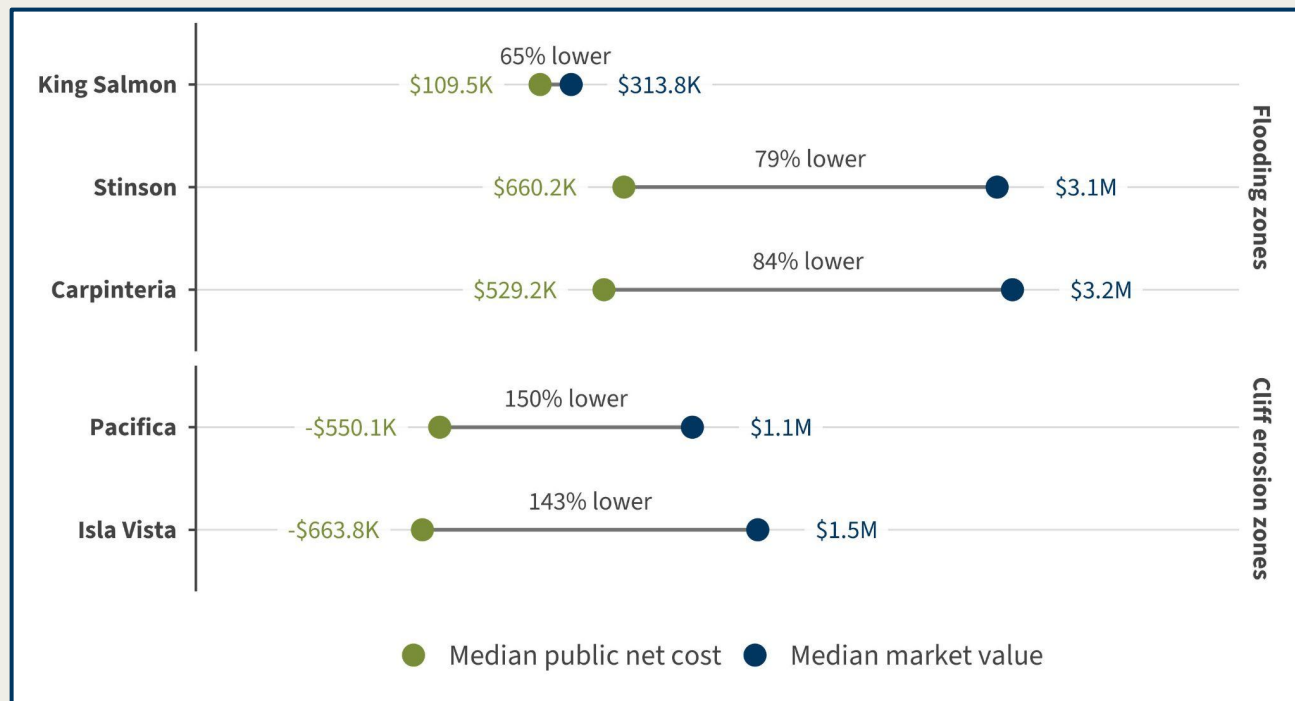
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Insights Across Case Studies

Buyout-leaseback
reduces public
retreat **costs** and
may **produce** net
public **benefit**



Insights Across Case Studies

Neighborhood-scale **buyout costs** are **comparable to other coastal adaptation** expenses.

**Buyout-Leaseback
in King Salmon**

\$6.5 M

**San Clemente Beach
Nourishment Project**

\$16.3 M

**Buyout-Leaseback
in Stinson**

\$132 M

**Stinson Bulkhead
Replacement**

\$295 M

Recommendations



**Build Community
Legitimacy**



**Test Feasibility
Through Pilots**



**Strengthen Policy
Foundations**

A coastal landscape featuring a sandy dune in the foreground covered with tall, dry, yellowish-brown grass. In the background, a dark blue ocean with white-capped waves meets a sandy beach. A few small figures of people are visible on the beach. The sky is a deep blue with some light, wispy clouds. The word "Questions?" is written in a large, white, serif font, centered over the middle of the image.

Questions?

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